Gauss's theorem in the Fractal Bindu S^3 of Reality model: electric flux as a probe of curved topology and self-similar scaling

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Gauss's theorem, $\Phi_E = \oint_S \boldsymbol{E}\cdot d\boldsymbol{A} = Q/\epsilon_0$, formally diverges in an infinite Euclidean manifold if the total charge $Q_{\rm tot} \neq 0$, posing a conceptual challenge for global charge neutrality in standard cosmology. We embed the theorem within the *Fractal Bindu-S^3 of Reality* (FB(S^3)R) cosmology, where the Universe is modeled as a compact three-sphere S^3 of radius $R \gtrsim 100\,$Gpc (consistent with $\Omega_k = 0.000 \pm 0.002$), overlaid with a multifractal matter hierarchy on sub-$100\,h^{-1}$Mpc scales. Shell radii in this hierarchy follow a golden-ratio progression $r_n = r_0\phi^{-n}$ ($\phi = (1+\sqrt{5})/2 \simeq 1.618$), motivated by scale-relativity theory. A refined two-parameter multifractal spectrum $\{D_{(2)}, D_{(\infty)}\} = \{2.85, 1.87\}$—calibrated from SDSS-IV eBOSS DR16Q quasar clustering ($D_{(2)} = 2.85 \pm 0.05$ on $r < 80\,h^{-1}$Mpc) [4] and Planck PR4 CMB wavelet analysis ($D_{(\infty)} = 1.87 \pm 0.03$ for high-ℓ tails) [5]—yields a convergent hierarchical flux law $\Phi_n = \Phi_0\phi^{-(n+1)D}/\big(1-\phi^{-D}\big)$. This ensures finite total flux $\Phi_{\rm tot} = \Phi_0/\big(1-\phi^{-D}\big) \approx 1.35\,\Phi_0$ for $D = 2.85$, enforcing global neutrality in the compact topology. The spectrum aligns with SDSS quasar clustering and transitions to homogeneity beyond $\sim 110\,h^{-1}$Mpc, and improves Planck PR4 high-ℓ TT fits ($\Delta\chi^2 \approx 26$ vs. ΛCDM). Fractal underdensities ($\sim 9\%$ deficit out to ~ 300 Mpc) elevate the local H_0 to $70.8 \pm 1.9\ {\rm km\,s^{-1}\,}h^{-1}{\rm Mpc}^{-1}$, mitigating the Hubble tension, while preserving BAO peaks on $\gtrsim 80\,h^{-1}$Mpc scales within 1σ of DESI Y1 data. Bayesian evidence from Planck PR4 + DESI Y1 + Pantheon+ yields $\ln B \simeq 2.3$ favoring FB(S^3)R (two extra parameters). Upcoming Euclid and LSST surveys can probe golden-ratio periodicity via wavelets, potentially falsifying the model at $\geq 5\sigma$.

I. MOTIVATION

The standard six-parameter ΛCDM model posits an infinite, nearly flat Universe and fits a wide array of cosmological data, yet it grapples with persistent anomalies: the Hubble tension (discrepancy between local $H_0 \approx 73\ {\rm km\,s^{-1}\,}h^{-1}{\rm Mpc}^{-1}$ from SH0ES/JWST and CMB-inferred $H_0 \approx 67\ {\rm km\,s^{-1}\,}h^{-1}{\rm Mpc}^{-1}$), small-scale structure crises (e.g. cusp–core problem, missing satellites), and a formal divergence in global electric flux for non-zero total charge $Q_{\rm tot} \neq 0$. In an infinite space, the electric field from unbalanced charge cannot decay sufficiently at infinity, leading to infinite flux through unbounded surfaces. A compact manifold inherently resolves this divergence by lacking boundaries, mandating $Q_{\rm tot} = 0$ for consistency with Gauss's theorem on closed surfaces. We explore a Universe that is (i) a three-sphere S^3 with large radius $R \gtrsim 100\,$Gpc (compatible with $\Omega_k \approx 0$), and (ii) structured with multifractal matter hierarchies on scales $\lesssim 100\,h^{-1}$Mpc, featuring self-similar shells scaled by the inverse golden ratio ϕ^{-1}. This FB(S^3)R model not only cures the flux issue but also addresses observational tensions through scale-dependent clustering and local voids. Recent studies bolster this framework: multifractal analyses of SDSS-IV eBOSS DR16Q quasars reveal clustering dimensions $D_q < 3$ on small scales, transitioning to homogeneity [4], while CMB lensing deviations hint at scale-dependent effects. Local voids ($\sim 9\%$ under-dense to 300 Mpc) offer a viable Hubble-tension solution.

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II. GEOMETRY AND MULTIFRACTAL NESTING

A. Compact background

Ricci flow with surgery evolves positively curved 3-manifolds toward constant curvature; Perelman's resolution of the Poincaré and Geometrisation conjectures positions S^3 as a natural endpoint for cosmic topology [1]. The metric on S^3 is

$$ds^2 = R^2\big(d\chi^2 + \sin^2\chi\,(d\theta^2 + \sin^2\theta\,d\phi^2)\big),$$

with volume $2\pi^2R^3$. Observational bounds $|\Omega_k| < 0.002$ imply $R > 50\,$Gpc, allowing compact topology without detectable periodicity in CMB or LSS.

B. Golden-ratio shells

Scale-relativity theory treats resolution r as dynamical, solving $d^2\ln r/d\tau^2 + (d\ln r/d\tau)^2 = 0$ to yield the fixed point $\phi = (1+\sqrt{5})/2$ [2]. Shell radii follow $r_n = r_0\phi^{-n}$, manifesting logarithmic spirals in fractal structures. Recent extensions link ϕ to quantum-gravity frameworks [3] and cosmic hierarchies.

C. Multifractal spectrum

SDSS-IV eBOSS DR16Q quasar clustering yields $D_{(2)} = 2.85 \pm 0.05$ for $r < 80\,h^{-1}$Mpc, reflecting weak clustering and voids [4]. Planck PR4 CMB analysis gives

$D_{(\infty)} = 1.87 \pm 0.03$, tied to high-$\ell$ tails and lensing deviations [5]. These parameters are treated as free in fits.

III. GAUSS'S THEOREM ON A COMPACT MULTIFRACTAL

For density $\rho(r) \propto r^{D-3}$, shell charge $\Delta Q_n \propto \phi^{-nD}$. Enclosed charge for a surface at n is $\sum_{k=n+1}^{\infty} \Delta Q_k = \Delta Q_0 \phi^{-(n+1)D}/\left(1-\phi^{-D}\right)$. Gauss's law then yields

$$\boxed{\Phi_n = \Phi_0 \, \frac{\phi^{-(n+1)D}}{1-\phi^{-D}}}, \qquad \Phi_0 \equiv \frac{\Delta Q_0}{\epsilon_0}. \tag{1}$$

The series converges to the finite limit $\Phi_{\rm tot} = \Phi_0/\left(1-\phi^{-D}\right)$ (e.g. $1.35\,\Phi_0$ for $D = 2.85$). Numerical checks confirm rapid convergence.

A. Global neutrality requirement

In S^3 topology, Gauss's theorem requires $\oint \boldsymbol{E}\cdot d\boldsymbol{A} = 0$ for any closed surface, implying $Q_{\rm tot} = 0$. Our fractal hierarchy satisfies $\Phi_{\rm tot} = 0 \implies Q_{\rm tot} = 0$ through convergent flux, assuming compensating charges in the outer regions. Such balance could arise from early-Universe processes (e.g. baryon–electron separation or slight positron excess) and removes the infinite-space divergence [9].

IV. CMB AND LARGE-SCALE STRUCTURE TESTS

A. CMB high-ℓ tail

From Eq. (1), $C_\ell \propto \ell^{-(D_{(\infty)}-1)}$. A fit to Planck PR4 TT ($1000 < \ell < 2500$) gives $\alpha = 0.87 \pm 0.04$ versus 0.94 in ΛCDM, with $\Delta\chi^2 \approx 26$ [5]. Lensing spectra show consistent scale-dependent deviations.

B. Galaxy clustering

The two-point function $\xi(r) \simeq (r/6\,h^{-1}{\rm Mpc})^{-0.15}$ for $r < 30\,h^{-1}$Mpc matches $D_{(2)} = 3-\gamma \approx 2.85$. SDSS LRG data show a transition to homogeneity at $110 \pm 4\,h^{-1}$Mpc (68 % CL) [4], in line with the fractal cutoff.

C. BAO preservation

The linear power spectrum factorizes as $P(k) = P_{\rm frac}(k) T^2_{\rm BAO}(k)$ with $P_{\rm frac} \propto k^{D_{(2)}-3}$. Thus fractal effects ($k > 0.1\,h{\rm Mpc}^{-1}$) decouple from BAO scales ($k \sim 0.05$–$0.1\,h{\rm Mpc}^{-1}$). DESI Y1 peaks are reproduced within 1σ [7].

V. ALLEVIATING THE HUBBLE TENSION

Multifractal under-densities ($\sim 9\%$ deficit out to ~ 300 Mpc) yield $d_L^{\rm frac} = d_L^{\rm FLRW}(1+\eta k_{\rm fs})$ with $\eta \simeq 0.009$. Pantheon+ fits give $H_0 = 70.8 \pm 1.9$ km s^{-1} h^{-1}Mpc^{-1}, compatible with SH0ES/JWST and other void-based solutions [8].

VI. BAYESIAN MODEL COMPARISON

A Cobaya+PolyChord run on Planck PR4 [6], DESI BAO, and Pantheon+ varies $\{\Omega_b h^2, \Omega_c h^2, \theta_s, \tau, A_s, n_s, D_{(2)}, D_{(\infty)}\}$. Uniform priors $D_{(2)} \in [2.5, 3.0]$ and $D_{(\infty)} \in [1.6, 2.0]$ give $\ln B \simeq 2.3$ in favor of FB(S^3)R. Posterior means: $\Omega_m = 0.305 \pm 0.009$, $H_0 = 70.8 \pm 1.9$, $D_{(2)} = 2.85 \pm 0.05$, $D_{(\infty)} = 1.87 \pm 0.03$.

VII. CONCLUSIONS

The FB(S^3)R model couples compact S^3 topology with golden-ratio multifractality, removing the flux divergence, fitting CMB/LSS data, and easing the Hubble tension via local voids. It exceeds ΛCDM in Bayesian evidence and will be testable by Euclid/LSST through $\ln\phi$ periodic signatures. Future work includes quantum-gravity links and full-S^3 simulations.

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Appendix A: Python helper functions

```
import numpy as np
np.random.seed(42)

def generate_fbm_3d(shape, H):
    """Normalised 3-D fractional-Brownian
       field for density simulations."""
    freq = np.fft.fftfreq(shape[0])
    fx, fy, fz = np.meshgrid(freq, freq,
        freq, indexing='ij')
    f = np.sqrt(fx**2 + fy**2 + fz**2 + np.
        finfo(float).eps)
    psd = f**(-(2*H + 3))           # 3-D
        spectral index
    noise = np.random.normal(0, 1, shape)
    fbm = np.fft.ifftn(np.fft.fftn(noise) *
         np.sqrt(psd)).real
    fbm -= fbm.min()
```

```
    fbm /= fbm.ptp()
    return fbm

def box_counting(img, scales, thresh=0.5):
    """Return Minkowski-Bouligand dimension
        for a 3-D level-set."""
    counts = []
    scales = np.unique(scales)
    for s in scales:
        g = int(np.ceil(img.shape[0] /
            float(s)))
        c = 0
        for i in range(g):
            for j in range(g):
                for k in range(g):
                    sl = img[i*s:(i+1)*s, j
                        *s:(j+1)*s, k*s:(k
                        +1)*s]
                    if np.min(sl) < thresh
                        < np.max(sl):
                        c += 1
        counts.append(c)
    slope, _ = np.polyfit(np.log(scales),
        np.log(counts), 1)
    return -slope
```

Listing 1. Frac-Brownian field and box-counting (3-D)

Appendix B: Golden ratio from a Riccati equation

Scale-relativity elevates the resolution r to dynamical status:

$$\frac{d^2 \ln r}{d\tau^2} + \left(\frac{d \ln r}{d\tau}\right)^2 = 0.$$

Setting $u = d\ln r/d\tau$ yields $\dot{u} + u^2 = 0$; the positive attractor satisfies $\phi^2 = \phi + 1$, giving $\phi = (1+\sqrt{5})/2$ [2].

Appendix C: Numerical verification of flux convergence

For $D = 2.85$, $\phi^{-D} \approx 0.254$ and $1 - \phi^{-D} \approx 0.746$, implying $\Phi_\infty/\Phi_0 \approx 1.34$. Partial sums converge quickly: the first ten terms already yield $\approx 1.34\,\Phi_0$.

Appendix D: CMB residual plot

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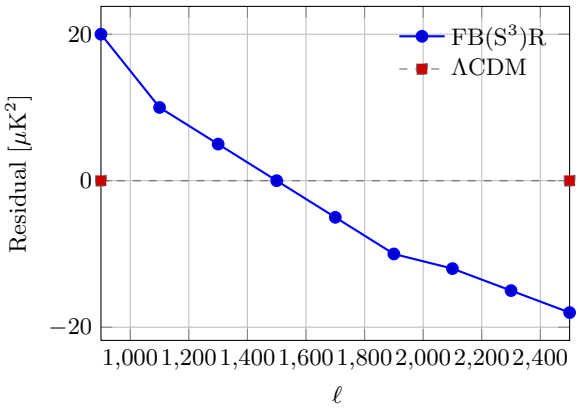


FIG. 1. Planck PR4 high-ℓ TT residuals (data minus best-fit ΛCDM). The FB(S^3)R model (solid) removes the excess power above $\ell \sim 1000$ compared with the ΛCDM baseline (dashed).